INCOME PREDICTION AND ANALYSIS USING R

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ABSTRACT

This paper examines the US census income dataset from Kaggle Machine learning Repository, using the same in the form of a web application that presents a user with useful insights into his/her future income depending upon their inputs. Our main objective is to find out whether the income of an individual would be more than 50000 per annum on the basis of few factors and attributes from the census data .We present the above mentioned information via a web application that is developed in the framework called Shiny, such that the user is able to use the model working at the back-end in a generalized form.

The census bureau is responsible for conducting the above surveys that leads to the formation of above dataset that contains millions of data points .Thus this rich dataset consist of large number of individuals whose dependent and independent attributes like age , workplace, education, marital status, occupation, relationship, race, sex, native country etc. are being taken in account .Hence the main aim is to find out unique insights and relating them to financial status of people of a particular country that would ultimately help in the prediction of economic and financial status.

Thus using these attributes, demographic graphs and plots are being made and classification of above data into different economic classes is being done using different classifiers such as logistic regression and Naïve Bayes model based classifiers.

General Terms

Framework, R, prediction, census data, and salary levels, demographic, economic classes, financial growth.

Keywords

Shiny framework, Logistic regression, Random forest, Education, Education num, Occupation, Relationship, Gender, Native country, Income.

1. INTRODUCTION

"A large income of a person is one of the best recipe for the happiness" is the famous quotes from English novelist Jane Austen. Income of a person is one of the most important factors that predict the standard and economic status of an individual. Thus, its importance and impact on nation's growth and development, providing meaningful insights which can serve as the main base for the decision making of nation's administration. The income census consists of 48,842 entries. Each one of them contains some information about an individual.

1. Education: the highest level of education that an individual has.
2. Education num: highest level of education as an integer. Integer should be greater than zero.
3. Age: age of a particular individual. In integer form it must be greater than zero.
4. Workclass : representation of employment status of a person i.e. private, govt. etc.
5. Occupation: general form of occupation that person has i.e. financial, technical tec.
6. Relationship: Represent the relationship of a person with other. That could be a husband, a father, son etc.

7. Gender: sex of an individual i.e. male or female.
8. Native country: to which a person belongs it is mainly where he was born.
9. Income: whether a person makes more than $50k per annum or not.

The dataset consist of 23.93% entries that are greater than 50k and rest 76.07% entries are smaller 50k.Wehave split the dataset into test sets while maintaining this distribution.

Many machine learning techniques are used to predict whether an observation makes greater than $50,000 a year or not. In similar manner as logistic regression, the original data set is split into a training set and a testing set. Models are now trained on the training set and validated on the testing set.

2. LITERATURE REVIEW

The income prediction was initially dependent on statistical models with the obvious constraint of limited observational ability. In the discussion paper, Prediction and Determination of Household Permanent Income [1], standard static welfare indicators (e.g. per capita expenditure and income) are concluded to be imperfect in the respect of determination of incomes as they typically contain a high transitory component. Framework which was employed was consistent is boosted with a causes equation where unobservable permanent income was specifically presented as a function of variables which played a key part in its determination. Simultaneous estimation of model helped to compare how well different obvious welfare indicators helps to identify permanent income but more importantly it enabled to predict permanent income using information contained in the causal variables and in the standard static welfare indicators.

Determination of income is primary application of predictive analytics where the customer segmentation takes place that is based on different demographical data. In the paper, “Comparative Analysis of Classification Models on Income Prediction” [2], it has been tried to identify this purpose using different classification techniques to reduce the risk and cost involved to predict certain income levels .Performance of each algorithm, identification of customers using classification techniques, has been demonstrated. In addition to this, an investigation analysis on true positives, false negatives, scored labels and scored probabilities has been provided.

Deep learning in recent times has brought breakthroughs in several areas, most recognizably voice and also in image recognition. In the work, “Deep learning applied to mobile phone data for Individual income classification “[3], deep learning has been extended into a new application domain - namely classification on the mobile phone datasets. The classic machine learning models have produced good results in the telecom prediction tasks, but it remains underutilized due to feature engineering which is resource - intensive and domain-specific. In addition to that, traditional machine learning algorithms requires separate feature engineering in the different countries. It has been demonstrated, how socio-economic status in large has de-identified mobile phone datasets can be successfully and accurately classified using deep learning, hence avoiding cumbersome and manual feature engineering process. A simple and an easy deep learning architecture have been implemented and are compared with traditional data mining models as the benchmark.

The paper, “Modeling and Predicting Individual Salaries: The Study of Finland's Unique Dataset “[4] that models wages by exploiting a unique Finnish dataset. Both of the genders, the data are divided into four income quartiles. Panel data model is estimated with three degree polynomials of age, duration of employment, Gross Domestic Product growth as explanatory variables. Many predictors have also been developed, for individual wages. Hence, this can be potentially useful information when designing and developing pension schemes.

3. WEB APPLICATION

On the front end, we develop an application that derives the data from our predictive models that work on the back end to provide relevant insights to the user.

For this purpose we use Shiny that is a package in R software. Shiny is an R package which makes it easy to build interactive web applications straight from R. One can host standalone apps on a webpage or embed them in R, markdown documents or build dashboards. One can also extend the Shiny application with CSS themes, html widgets, and JavaScript actions.

We need to integrate this application with a set of statistical inferences from the dataset we have after analyzing and exploring the different independent variables. Some of these inferences are shown below:

After exploring the age variable, we get the following boxplot (Fig.1):

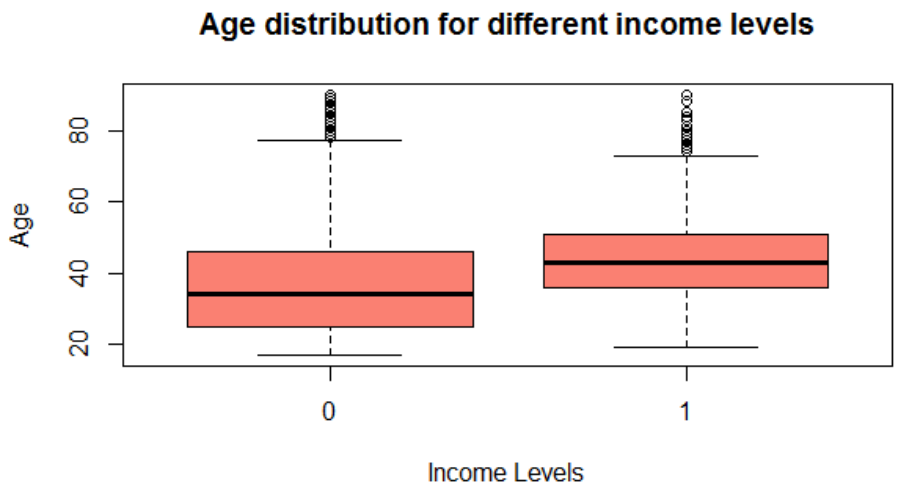

Fig.1. Age v/s income levels boxplot

Hence, we need a method to integrate the analysis with the prediction from our predicted model. Here is where Shiny framework comes into play. The final application runs as an R script with an R session running in the background. Some of the snapshots showing the entire analysis and the resulting output are given in (Fig 2-6)

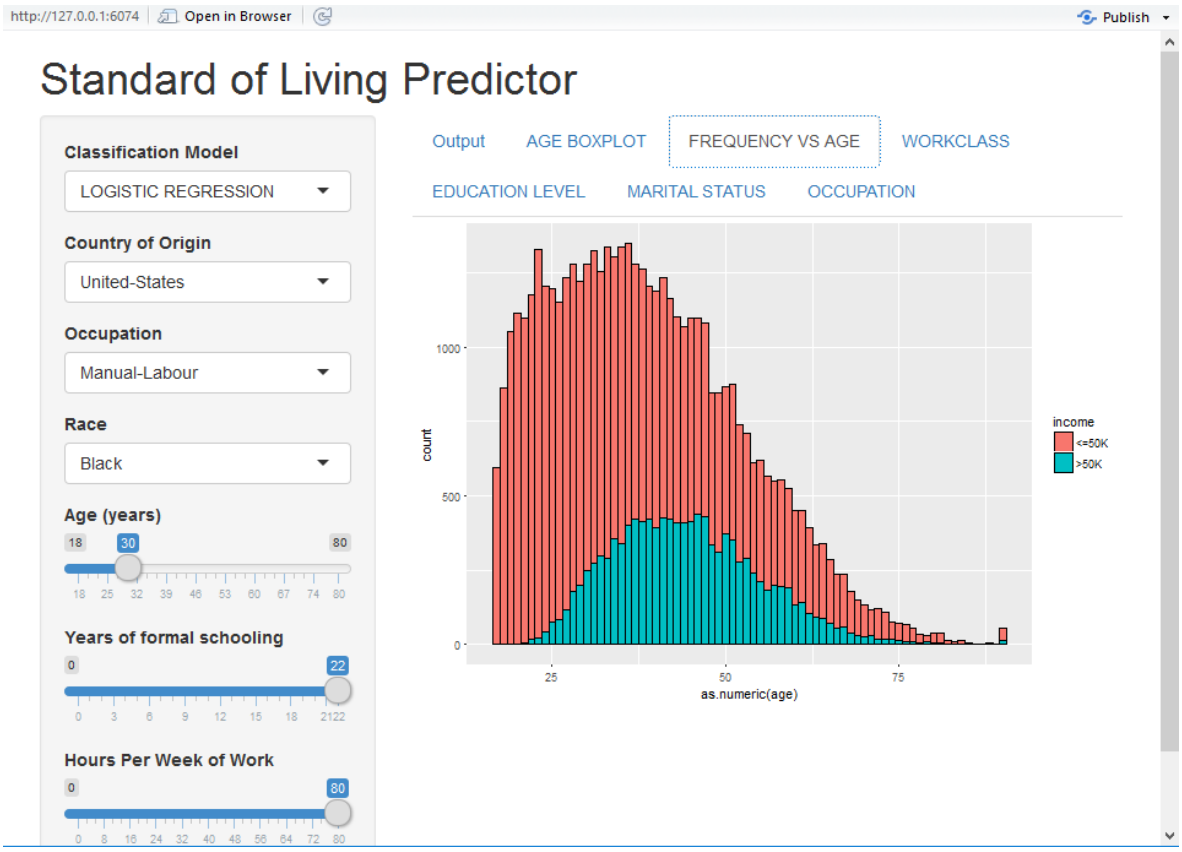

Fig.2. Frequency v/s age tab

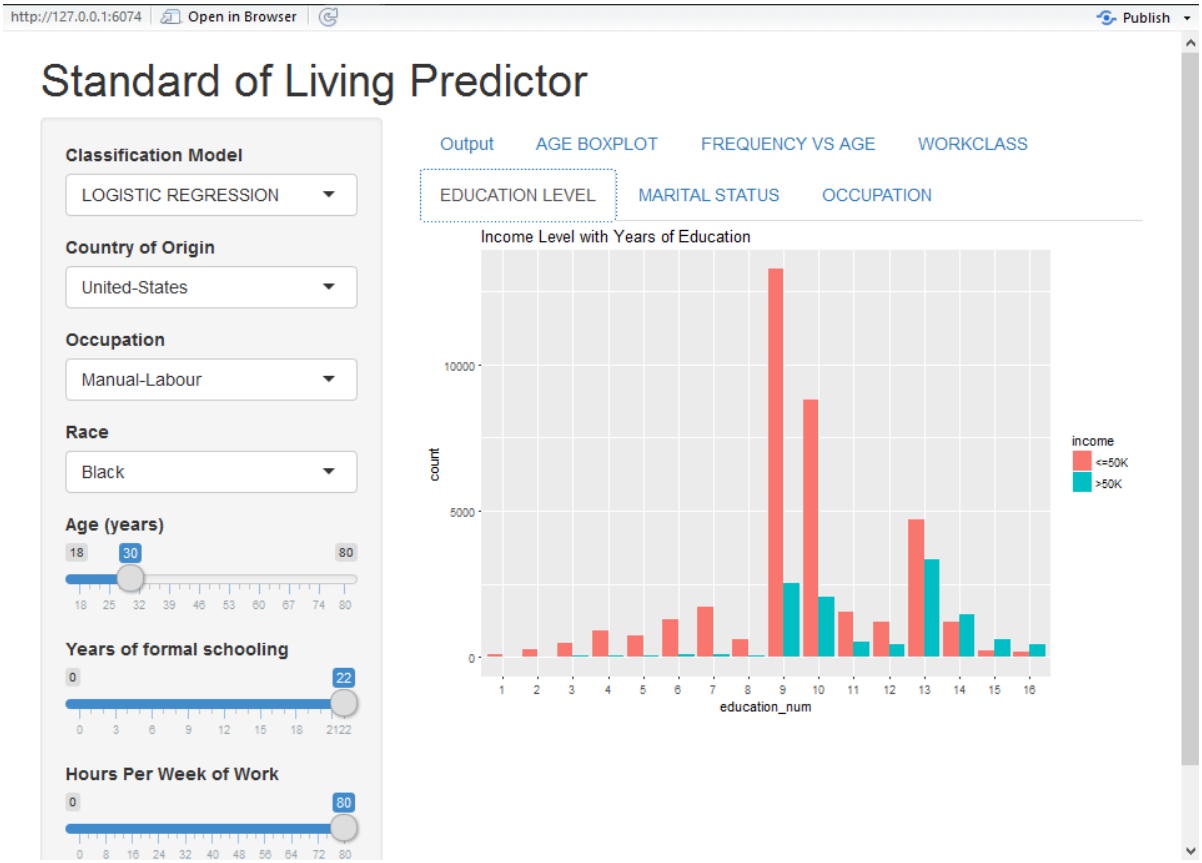

Fig.3. Education Level Tab

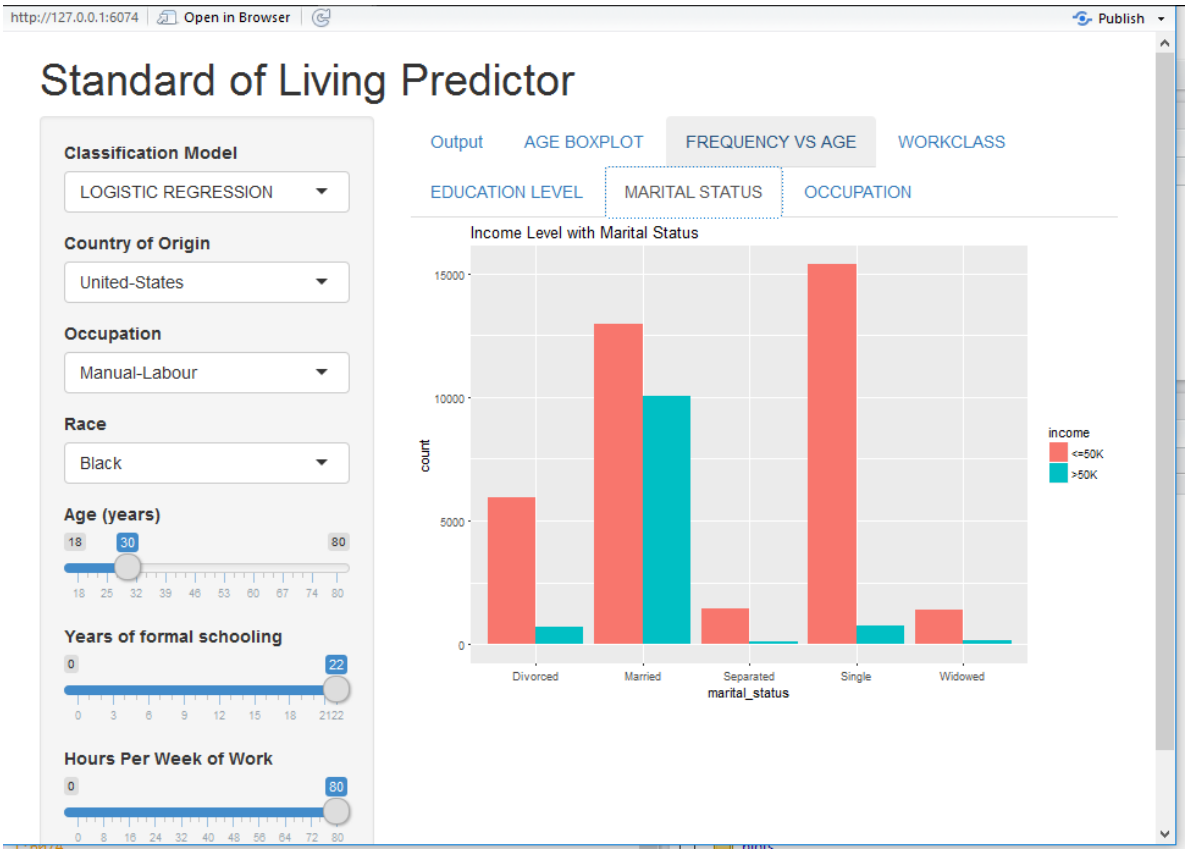

Fig 4.Marital Status tab

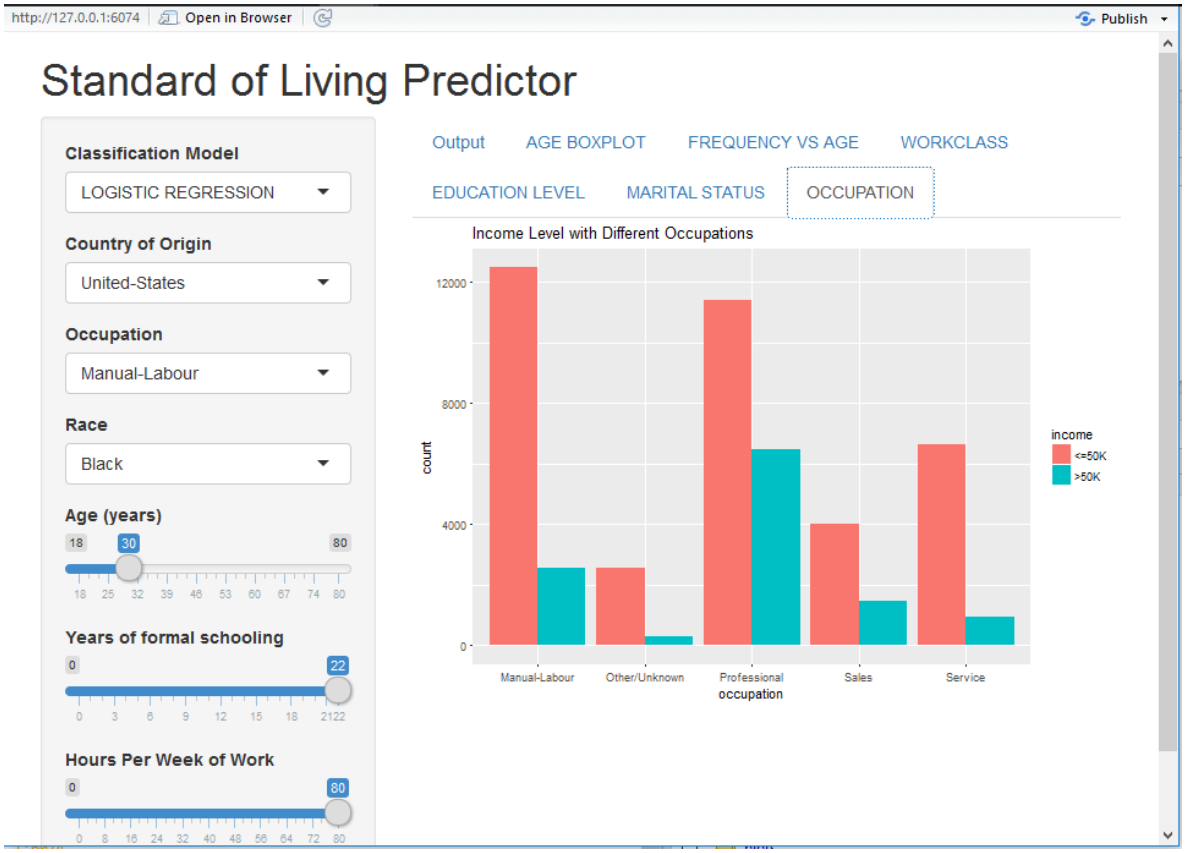

Fig 5 Occupation Tab

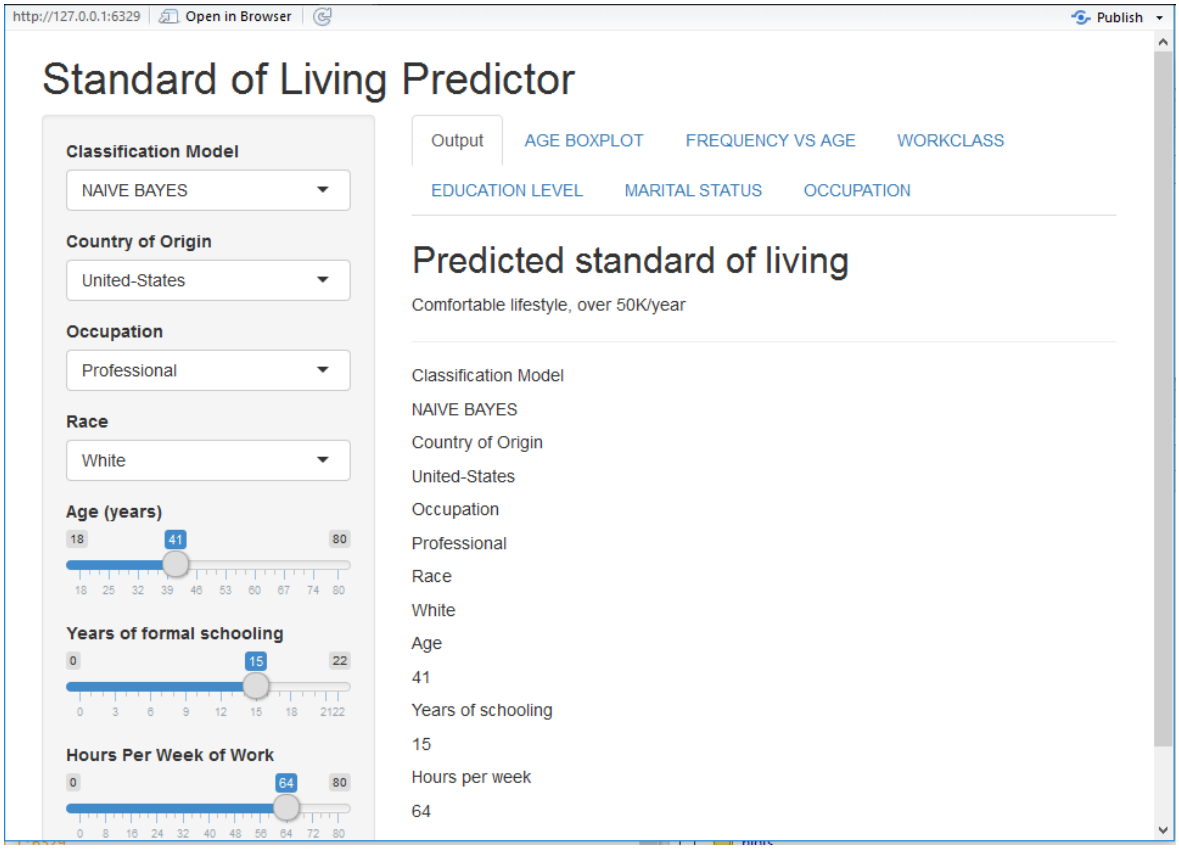

Fig.6 Output Tab

4. PREDICTIVE MODEL

This is the model supposed to be working at the backend of our web application that is to be developed in the above mentioned frame work (Shiny).

4.1 Base lining the Data

As mentioned before, we see a distribution that is approximately 24 % which is greater than 50k and rest 76% is smaller than 50k .For the base lining of data we take in consideration the majority of data that is 76% which is smaller than 50k label.

4.2 Logistic Regression

It is a model where dependent variables i.e. DV is categorical .This consist of binary dependent variables that is it can take only 2 values that is only 1 or 0.whicvh represents the data results to be true or false , pass or fail , alive or dead etc. .There are cases in which there are more than 2 outcomes, categories maybe taken in account in multinomial logistic regression, or if there are multiple categories are present or ordered then we can take ordinal logistic regression .Logistic regression is an example of discrete choice model, in economics.

David cox was developer of logistic regression in 1958.Binary logistic model is used to predict data based on one or more predictor based on binary values .Presence of risk factor increases the odds of an outcome by a specified number.

Latent variable interpretation: logistic regression can be taken as one in which we have to find beta parameter .$y = \beta 0 + \beta 1x + \varepsilon$, where ε is the error distribution .If the value of above equation i.e. $\beta 0 + \beta 1x + \varepsilon > 0$ then 1 else 0. In associate latent variable i.e. $y' = \beta 0 + \beta 1x + \varepsilon$, error is not taken into consideration and y' is also unobservable hence called as latent that is data depends on x and y.

Logistic regression is based on the measuring the relationships between the dependent variables, it treats the same values as probit regression using similar techniques where as other use cumulative normal distribution curve. In Latent variable interpretation, first one assumes a standard logistic distribution of errors and second one standard normal distribution of errors.

The figure (Fig.7) given below provides the relation between various independent variables and the probability of the dependent variable, that is, the income of the particular person to be in class that represents people who are earning more than 50000 dollars or those who earn less, which is determined by observing whether probability lies closer to 1 or 0.

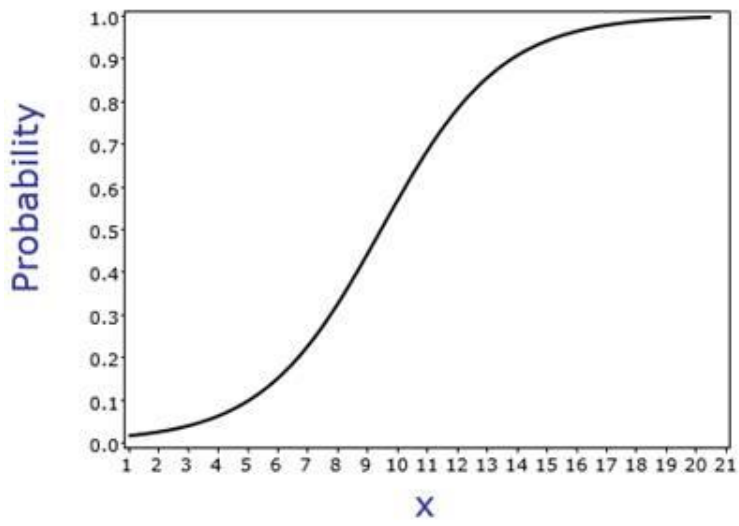

Fig.7. Logistical Sigmoid function

4.3 Random Forests

Random forests or random decision forests is an effective learning method which is used for classification, regression and other such machine learning and data science problems, that operates by construction of multiple decision trees at training time and finding the output of the class i.e. the mode of the classes or classification or mean prediction (regression) of the individual tree. Random decision forests correct the decision trees' habit of over fitting to their respective training set.

The algorithm for random decision forests was first created by Tin Kam Ho, by making use of the random subspace method, that, in Ho's formulation, it is one of the way to implement the "stochastic discrimination" approach for the classification proposed by Eugene Kleinberg. Although random forest algorithms have been inherently designed to work only with multidimensional data, it has been shown that one can also use them for arbitrary objects (like graphs or time series) with the use of only pairwise similarities between objects. This variant of the algorithm is referred to as a similarity forest.

Tin Ho in 1995 established that forests of trees splitting with oblique hyper planes, if randomly restricted to be sensitive to only selected feature dimensions, can gain accuracy as they grow without the factor of overtraining. Further work along the same lines concluded that other splitting methods behave similarly. To be noted that this observation of a more complex classifier (a larger forest) getting more accurate nearly monotonically is in the sharp contrast to the common notion that the complexity of a classifier can only grow to a certain level of accuracy before being affected by over-fitting. The explanation of this forest method's resistance to overtraining can be found in Kleinberg's theory of stochastic discrimination.

In the early stages of development of Breiman's notion of random forests was influenced by the work of Amit and Geman, who had introduced the idea of searching over random subset of the available decisions when splitting a node, in the context of growing a single tree. Idea of random subspace selection from Ho was also influential in the designing of random forests. In this method, forest containing trees is grown and variation among the trees is then introduced by projecting the training data into a randomly chosen subspace before fitting each tree or each node. Finally, idea of randomized node optimization, where the decision at each of the node is selected by a randomized procedure, rather than a deterministic optimization was first introduced by Dietterich.

Similarly, other classifiers such as Naïve Bayes may also be used.

5. CONCLUSIONS

In the project we have created a web application that makes a user's experience of interactions with our underlying predictive model easier and more comprehensive. We have focused on the modeling and prediction of future salaries of individuals using their past salaries. The main features of this analysis were a dataset of individual careers that is obtained from United States census. Various individual and economy attributes have been taken in consideration to estimate a general model for the analysis which could be further used in different fields. The main objective is not to present all the necessary details of the US data, but is the demonstration of the main features and opportunities of panel models that is based on high-quality data. Modeling is never conclusive in itself. Here the interest has been in predicting for planning purposes to further causes that prove beneficial for one's career. Wage forecasts may also provide critical information about the pension to come and regarding the individual risk

In this project, we constructed a model that is able to predict, with considerable accuracy, whether an individual will earn over $50k/yr. based on various data like marital status, education, age or nationality. The model may be tuned to meet our needs which could be to classify high-income customers more accurately and allow the users of our application to chart out their career paths in a better way.

6. ACKNOWLEDGMENTS

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